Master of Business Analytics

Prescriptive Analytics & Optimization

MBAN 5540

Class 06: Network Models

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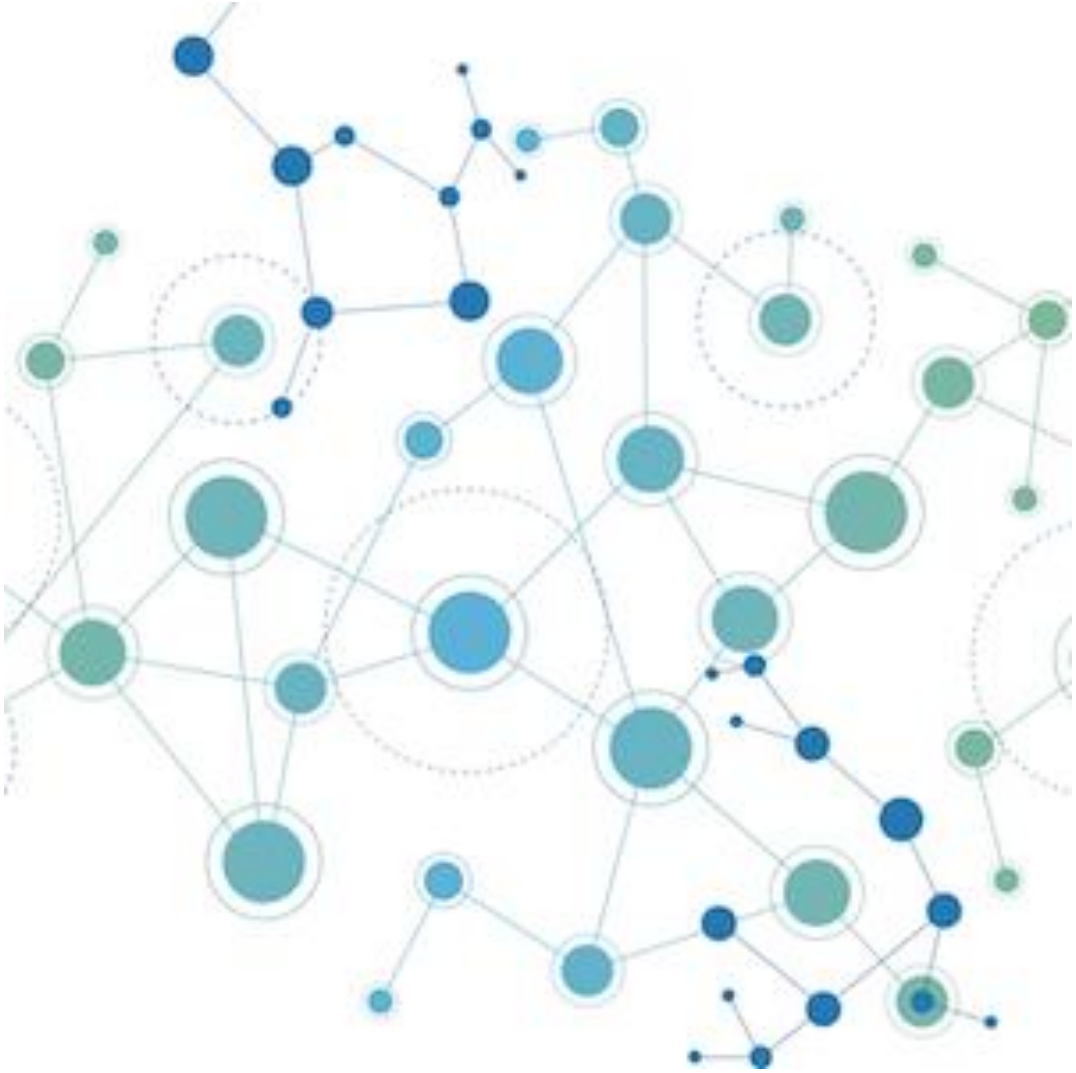
Network Models

- The use of networks helps in visualizing and understanding the managerial problem.

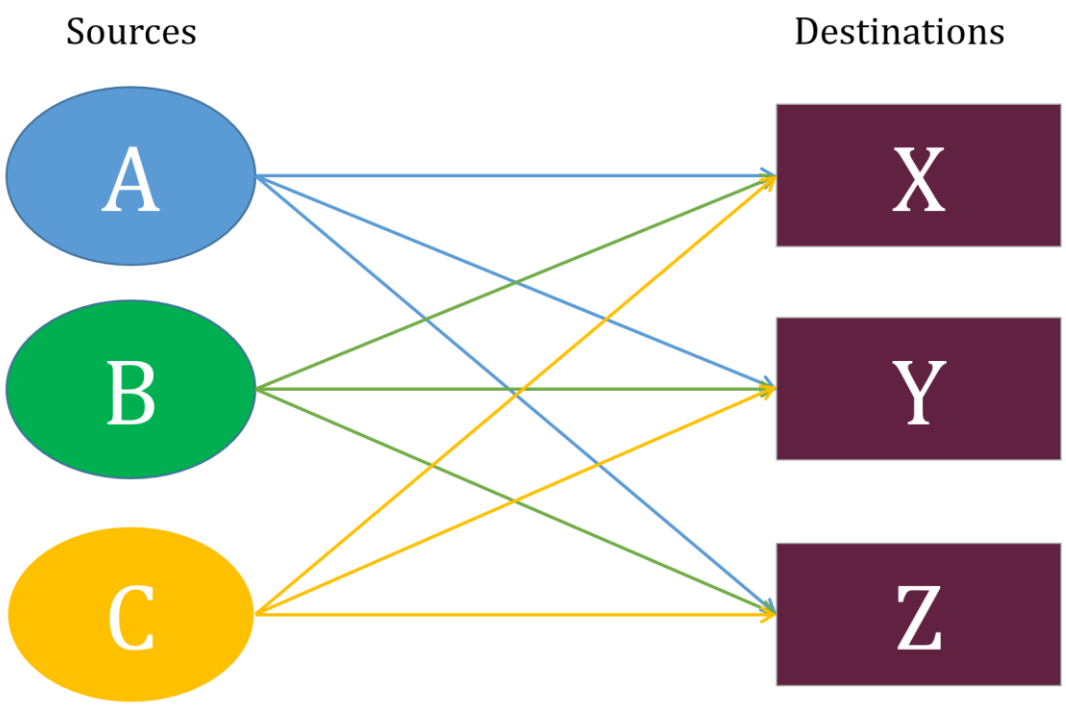
- The points on the network are referred to as **nodes** and the lines on the network that connect these nodes are called **arcs**.



Transportation Problem

- The basic transportation problem is concerned with finding the best (usually least cost) ways to distribute goods from **sources** such as factories to final **destination** such as retail outlets.

- The transportation problem deals with distribution of goods from several points of supply (origins or sources) to a number of points of demand (destination)
- Usually, we are given a capacity of goods at each source, requirement for goods at the destination, and the shipping cost per unit from each source to each destination.



Transportation Problem

Transportation Problem

$$
\begin{aligned}
\text{Minimize} \quad & \sum_{i=1}^{m} \sum_{j=1}^{n} c_{ij} x_{ij} \\
\text{Subject to} \quad & \sum_{j=1}^{n} x_{ij} \leq S_i && (i = 1, 2, \ldots, m) \\
& \sum_{i=1}^{m} x_{ij} \geq D_j && (j = 1, 2, \ldots, n) \\
& x_{ij} \geq 0 && (i = 1, 2, \ldots, m, j = 1, 2, \ldots, n)
\end{aligned}
$$

Parameters

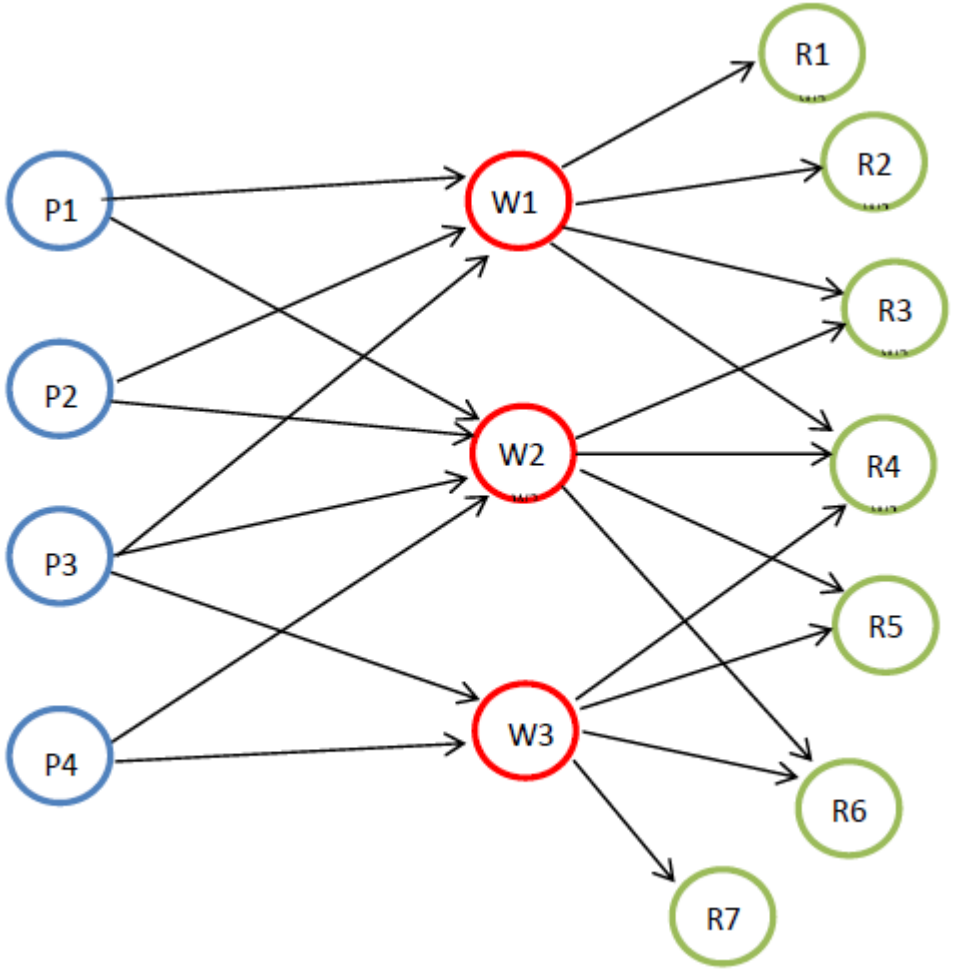
- m and n are the number of supply and demand locations respectively.
- c_{ij} is the cost of shipping one unit of goods from supply location i to demand location j.
- S_i is the number of units of goods that can be shipped from supply location i.
- D_j is the number of units of goods that need to be shippedto demand location j.

Decision Variables

- x_{ij}: the number of units of goods shipped from supply location i to demand location j.

Transshipment Problem

- It there are intermediate points such as regional distribution centres, where the items must go before being shaped to the final destination, then the transportation problem becomes a transshipment problem.



Transshipment Problem

Transhipment Problem

$$
\begin{aligned}
\text{Minimize} \quad & \sum_{i=1}^{m}\sum_{k=1}^{t} c_{ik}x_{ik} + \sum_{k=1}^{t}\sum_{j=1}^{n} c_{kj}x_{kj} \\
\text{Subject to} \quad & \sum_{k=1}^{t} x_{ik} \le S_i && (i = 1, 2, \dots, m) \\
& \sum_{i=1}^{m} x_{ik} = \sum_{j=1}^{n} x_{kj} && (k = 1, 2, \dots, t) \\
& \sum_{k=1}^{t} x_{kj} \ge D_j && (j = 1, 2, \dots, n) \\
& x_{ik}, x_{kj} \ge 0 && (i = 1, 2, \dots, m, j = 1, 2, \dots, n, k = 1, 2, \dots, t)
\end{aligned}
$$

Parameters

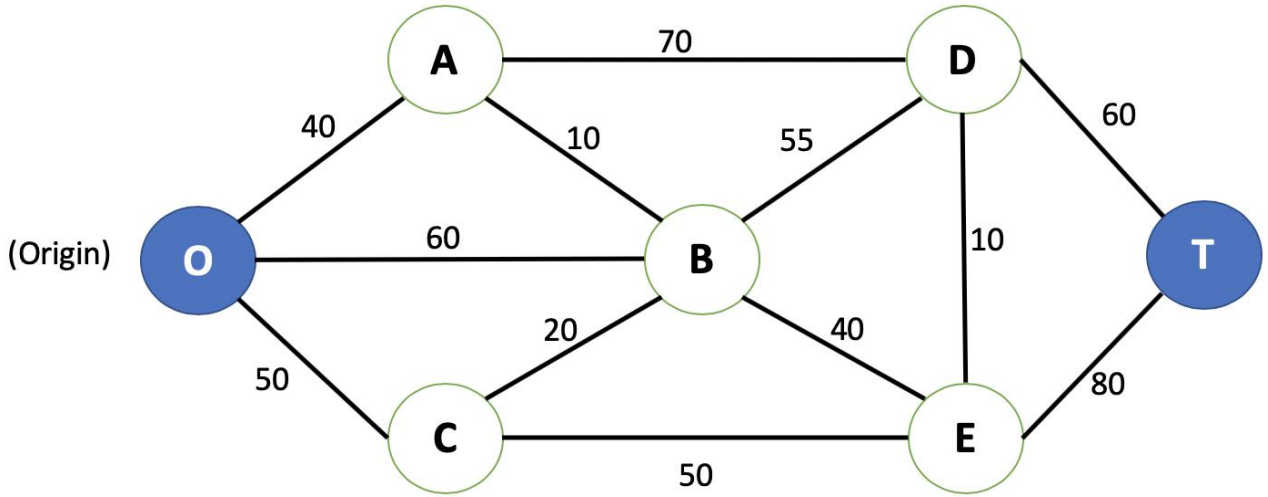
- m is the number of supply locations.
- t is the number of transshipment locations.
- n is the number of demand locations.
- c_{ik} is the cost of shipping one unit of goods from supply location i to transshipment location k.
- c_{kj} is the cost of shipping one unit of goods from transshipment location k to demand location j.
- S_i is the number of units of goods that can be shipped from supply location i.
- D_j is the number of units of goods that need to be shippedto demand location j.

Decision Variables

- x_{ik}: the number of units of goods shipped from supply location i to transshipment location k.
- x_{kj}: the number of units of goods shipped from transshipment location k. to demand location j.

Shortest Path Problem

- You can consider a problem in which the objective is to determine the shortest path or route in a network.



Shortest Path Problem

Shortest Path Problem

$$
\begin{aligned}
\min \quad & \sum_{(i,j)\in A} c_{ij}x_{ij} & \\
\text{S.t.:} \quad & \sum_{(i,j)\in A} x_{ij} = 1 & (i \text{ is the origin node}) \\
& \sum_{(i,j)\in A} x_{ij} = \sum_{(i,j)\in A} x_{ji} & (\text{all transshipment nodes}) \\
& \sum_{(i,j)\in A} x_{ij} = 1 & (j \text{ is the destination node}) \\
& x_{ij} \in \{0,1\} & \forall (i,j) \in A
\end{aligned}
$$

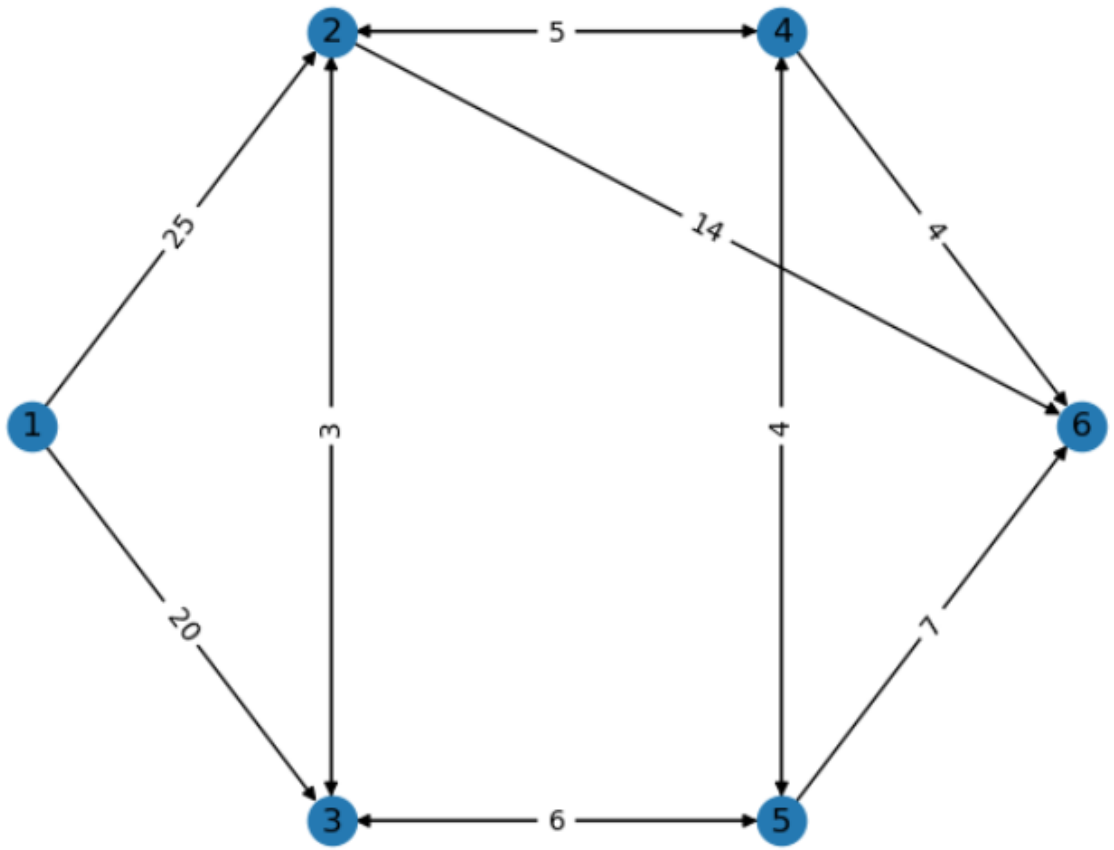
Parameters

- A is the set of Arcs, $(i,j) \in A$
- c_{ij} is the distance betweeen nodes i and j

Decision Variables

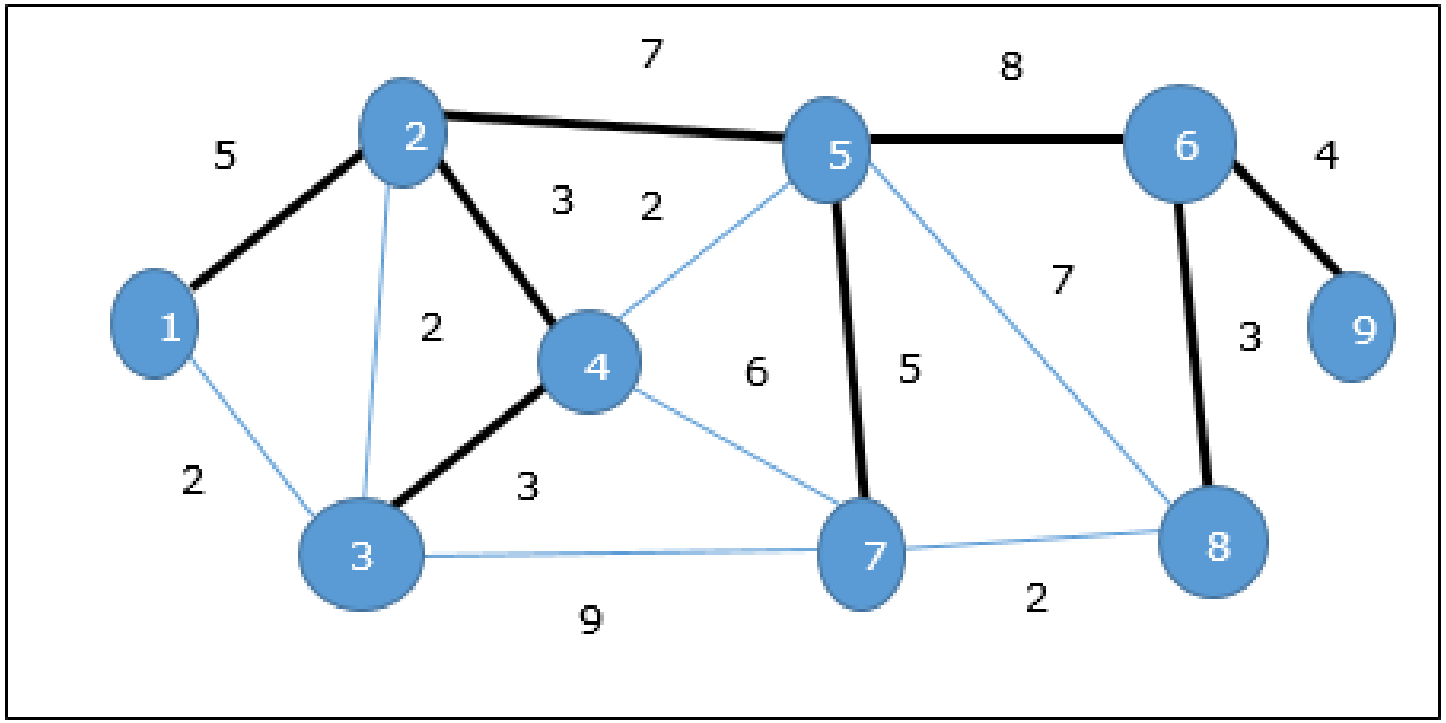
- $x_{ij} = \begin{cases} 1, & \text{if arc } (i,j) \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$

Shortest Path Problem - Example



Spanning Trees

- A tree is a *connected* graph that has no *cycles*



Minimum Spanning Tree

- The objective is to find the minimum cost tree so that all the nodes are connected.

Minimum Spanning Tree Problem

$$
\begin{aligned}
\min \quad & \sum_{(i,j)\in E} c_{ij}x_{ij} \\
\text{S.t.:} \quad & \sum_{(i,j)\in E} x_{ij} = n-1 \\
& \sum_{\substack{(i,j)\in E \\ i\in S; j\in S}} x_{ij} \le |S|-1 \qquad \forall S \subset V\ (S \neq \emptyset) \\
& x_{ij} \in \{0,1\} \qquad \forall (i,j) \in E
\end{aligned}
$$

Parameters

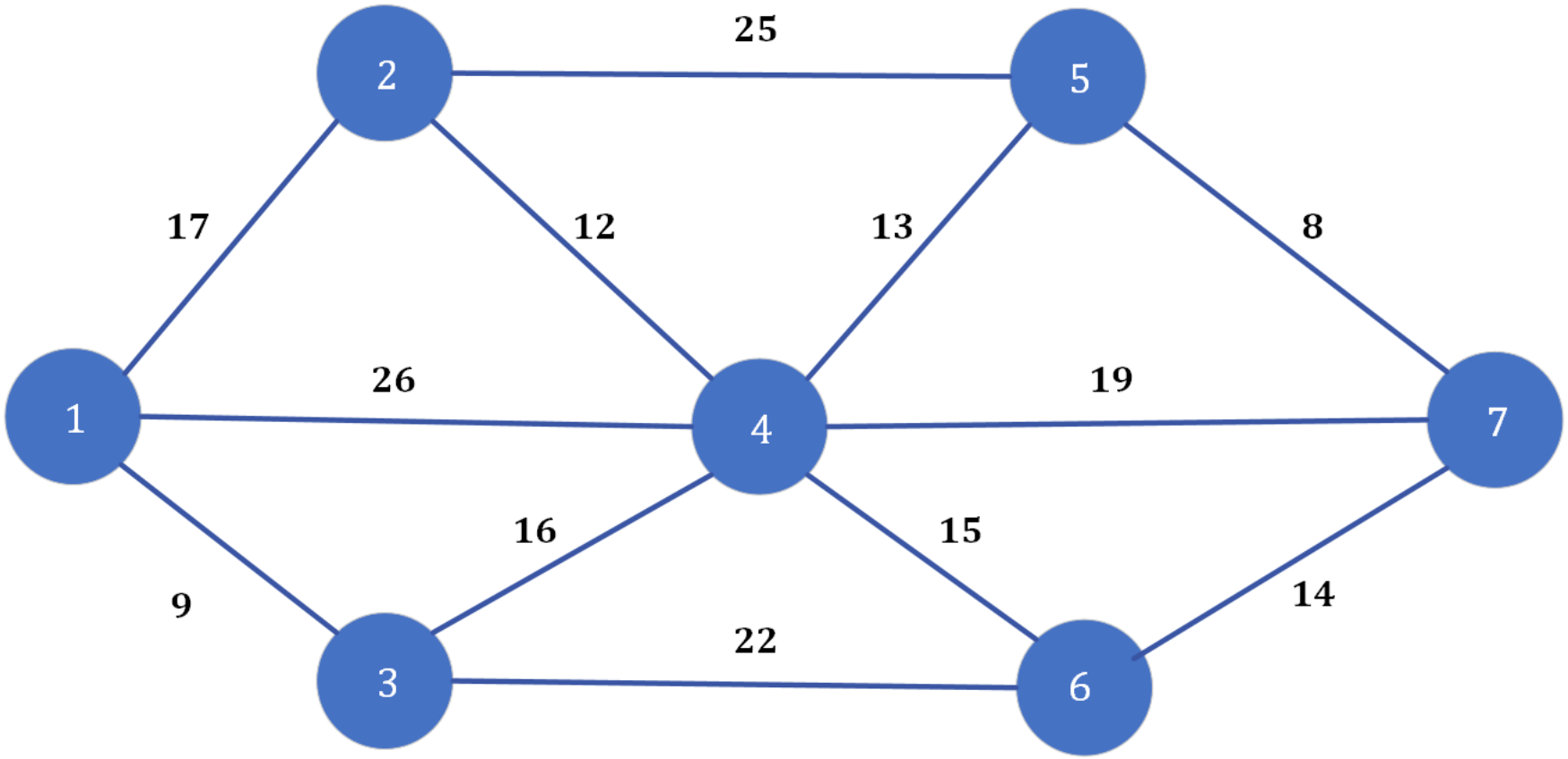
- $G = (V, E)$ is the graph; V is the set of nodes ($|V| = n$) and E is the set of arcs (edges)
- c_{ij} is the distance betweeen nodes i and j

Decision Variables

- $x_{ij} = \begin{cases} 1, & \text{if arc } (i,j) \text{ is selected} \\ 0, & \text{otherwise} \end{cases}$

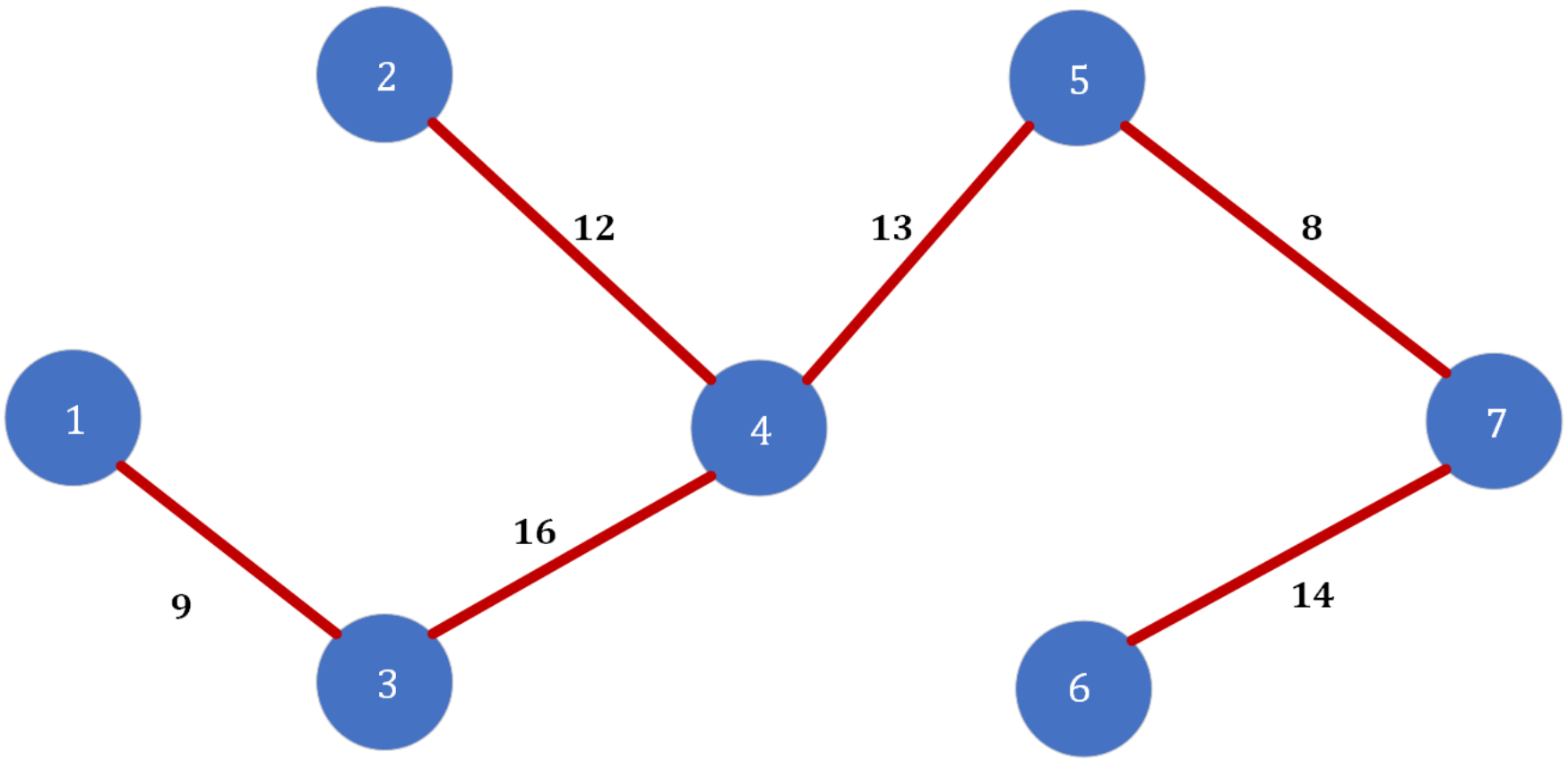
Minimum Spanning Tree

- The minimum spanning tree can be easily solved using a **greedy** approach.



Minimum Spanning Tree

- Min Cost = $8 + 9 + 12 + 13 + 14 + 16 = 72$



Greedy Approach and Optimality

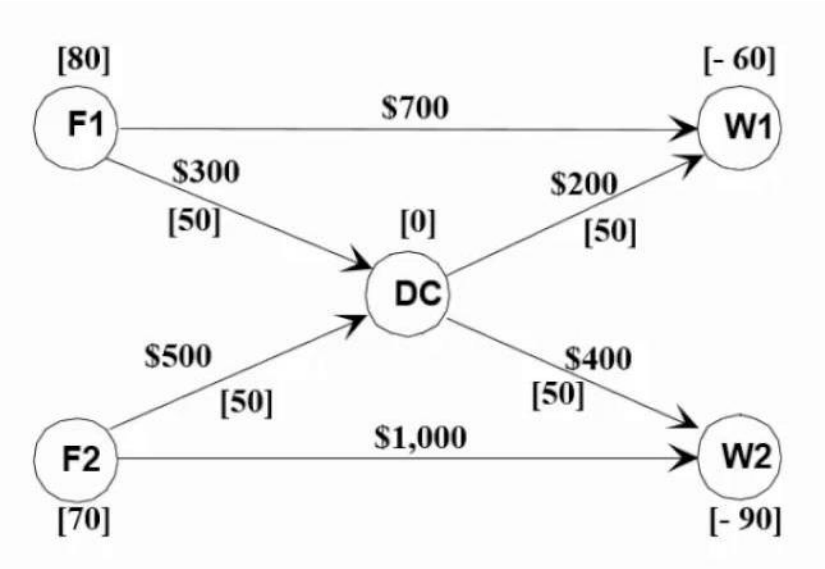
- The greedy algorithms are *not guaranteed* to find the optimal solution.

- Example: Find the minimum number of items from the below set of weights to fill a knapsack with 50 kg capacity

- Suppose we have unlimited number of each item:

$$\{1, 25, 35\}$$

Minimum Cost Flow Problems



Each arc has a capacity and a flow cost

Each node has a supply (+) or demand (-).

Sum of supply for all supply nodes = Sum of demand for all demand nodes

The objective is to find the cost of the flow.

Minimum Cost Flow Problems

Minimum Cost Flow Problem

$$
\begin{aligned}
\min \quad & \sum_{(i,j)\in E} c_{ij} x_{ij} \\
\text{S.t.:} \quad & \sum_{(i,j)\in E} x_{ij} - \sum_{(j,i)\in E} x_{ji} = b_i && \forall i \in V \\
& x_{ij} \leq u_{ij} && \forall (i,j) \in E \\
& x_{ij} \in \mathbb{R}^+ && \forall (i,j) \in E
\end{aligned}
$$

Parameters

- $G = (V, E)$ is the graph; V is the set of nodes ($|V| = n$) and E is the set of arcs (edges)
- c_{ij} is the cost of flow on arc (i, j)
- u_{ij} is the capacity of arc (i, j)
- $b_i = \begin{cases} \text{Supply of node } i, & \text{if } b_i \text{ is a positive number} \\ \text{Demand of node } i, & \text{if } b_i \text{ is a negative number} \end{cases}$

Decision Variables

- x_{ij} is the amount of flow on arc (i, j)

Minimum Cost Flow Problems

- What happens if supply/demand is zero for all nodes except for two and capacities are unlimited:

$$b_i = \begin{cases} 1, & i = \text{Origin} \\ -1, & i = \text{Destination} \\ 0, & \text{otherwise} \end{cases} \qquad u_{ij} = \infty, \ \forall (i,j) \in E$$

- The problem will become a shortest path problem!